

MATH120-24S2 Discrete Mathematics

Assignment 2

1. Let p and q denote the following statements:

p : *Bella is a cat.*

q : *Bella has teeth.*

Consider the argument

$$p \rightarrow q.$$

We assume that this is a valid argument (that is, that cats have teeth).

- (a) Write down the contrapositive of the statement in English.
- (b) Write down the inverse of the statement in English.
- (c) Write down the converse of the statement in English and give a (real) example which shows that the inverse is not necessarily a valid argument.

Solution:

- (a) If Bella does not have teeth, then Bella is not a cat.
- (b) If Bella is not a cat, then Bella does not have teeth.
- (c) If Bella has teeth, then Bella is a cat. The inverse was: If Bella is not a cat, then Bella does not have teeth. This is not a valid argument. A counter-example would be a girl (or rabbit, or cow etc...) named Bella; in that case, Bella is not a cat and Bella has teeth. *Note that this question was meant to ask to show that the converse is not necessarily a valid argument. However, to show that the converse isn't necessarily a valid argument, the girl named Bella is still a counterexample: Bella has teeth and Bella is not a cat. The fact that this is a counterexample for both the converse and the inverse isn't surprising: we have seen that the inverse and converse are logically equivalent to each other.*

2. Consider the argument

$$((p \vee q) \wedge (r \rightarrow q) \wedge (q \rightarrow \neg r)) \rightarrow p.$$

(a) Determine whether it is a valid argument by means of a truth table. Which rows of the table are crucial for assessing the validity of the argument and which rows can be ignored?

(b) Is the statement a logical implication? In other words, does

$$((p \vee q) \wedge (r \rightarrow q) \wedge (q \rightarrow \neg r)) \Rightarrow p?$$

Why/why not?

Solution:

(a)

p	q	r	$p \vee q$	$r \rightarrow q$	$\neg r$	$q \rightarrow \neg r$	$(p \vee q) \wedge (r \rightarrow q) \wedge (q \rightarrow \neg r)$	$((p \vee q) \wedge (r \rightarrow q) \wedge (q \rightarrow \neg r)) \rightarrow p$
0	0	0	0	1	1	1	0	1
0	0	1	0	0	0	1	0	1
0	1	0	1	1	1	1	1	0
0	1	1	1	1	0	0	0	1
1	0	0	1	1	1	1	1	1
1	0	1	1	0	0	1	0	1
1	1	0	1	1	1	1	1	1
1	1	1	1	1	0	0	0	1

There are two possible ways to go about this: either we consider only those rows for which the premise is true, and check whether in these cases the conclusion is true as well (these are rows 3,5,7). The other way is to only consider those rows for which the conclusion is false, and then check whether for all of those the premise is false as well (these are rows 1,2,3,4).

(b) Looking at the 1 in **bold** in the truth table above, we see that in the case that p and r are false while q is true, the truth value of $(p \vee q) \wedge (r \rightarrow q) \wedge (q \rightarrow \neg r)$ is 1, but the truth value of p is 0. Consequently (and by definition),

$$((p \vee q) \wedge (r \rightarrow q) \wedge (q \rightarrow \neg r)) \not\Rightarrow p.$$

3. (a) Prove that the statement is true or give a counterexample that disproves it: $\forall n$, if n is a positive integer then $n^2(n-6) = 6-11n$.
- (b) Prove that the statement is true or give a counterexample that disproves it: $\forall n$, if n is an even integer then $11n$ is even.

- (c) We asked ChatGPT the same question about the statement $\forall n$, if n is an integer, then either $n+1$ is even or n^2 is even. We received:
To prove or disprove $\forall n$, if n is an integer, then either $n+1$ or n^2 is even:

Let's attempt to disprove this statement by finding a counterexample. If n is an odd integer, then $n = 2k+1$ for some integer k . So, $n+1 = 2k+1+1 = 2k+2 = 2(k+1)$, which is even. However, $n^2 = (2k+1)^2 = 4k^2+4k+1 = 2(2k^2+2k)+1$, which is odd. So, for odd integers, neither $n+1$ nor n^2 is even. Thus, the statement is false. Explain why this answer is wrong and what needs to be done to fix it.

Solution:

- (a) False. As a counter-example, consider $n = 4$. Then $n^2(n-6) = -64$ but $6-11n = -38$.
- (b) True. *Proof:* Let n be an even integer. Then by definition, there exists an integer k such that $n = 2k$. Then

$$11n = 11(2k) = 22k = 2(11k)$$

Since $11k$ is an integer, $11n$ is even.

- (c) ChatGPT correctly proves that if n is odd, then $n+1$ is even, and that if n is odd, n^2 is odd. It then incorrectly claims that if n is odd, neither of $n+1$ nor n^2 is even, which is obviously false. The statement itself is correct. We already have the statement is true for n is odd, so we now assume that n is even. In that case, $n = 2k$ for some integer k , and we find that $n^2 = 4k^2 = 2(2k^2)$. Since $2k^2$ is an integer, this means that n^2 is even. We conclude that both in the case that n is odd as in the case that n is even, the statement holds. Since an integer is either even or odd, this proves the statement.